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Aether Intelligent Messaging (AIM): Hands-On

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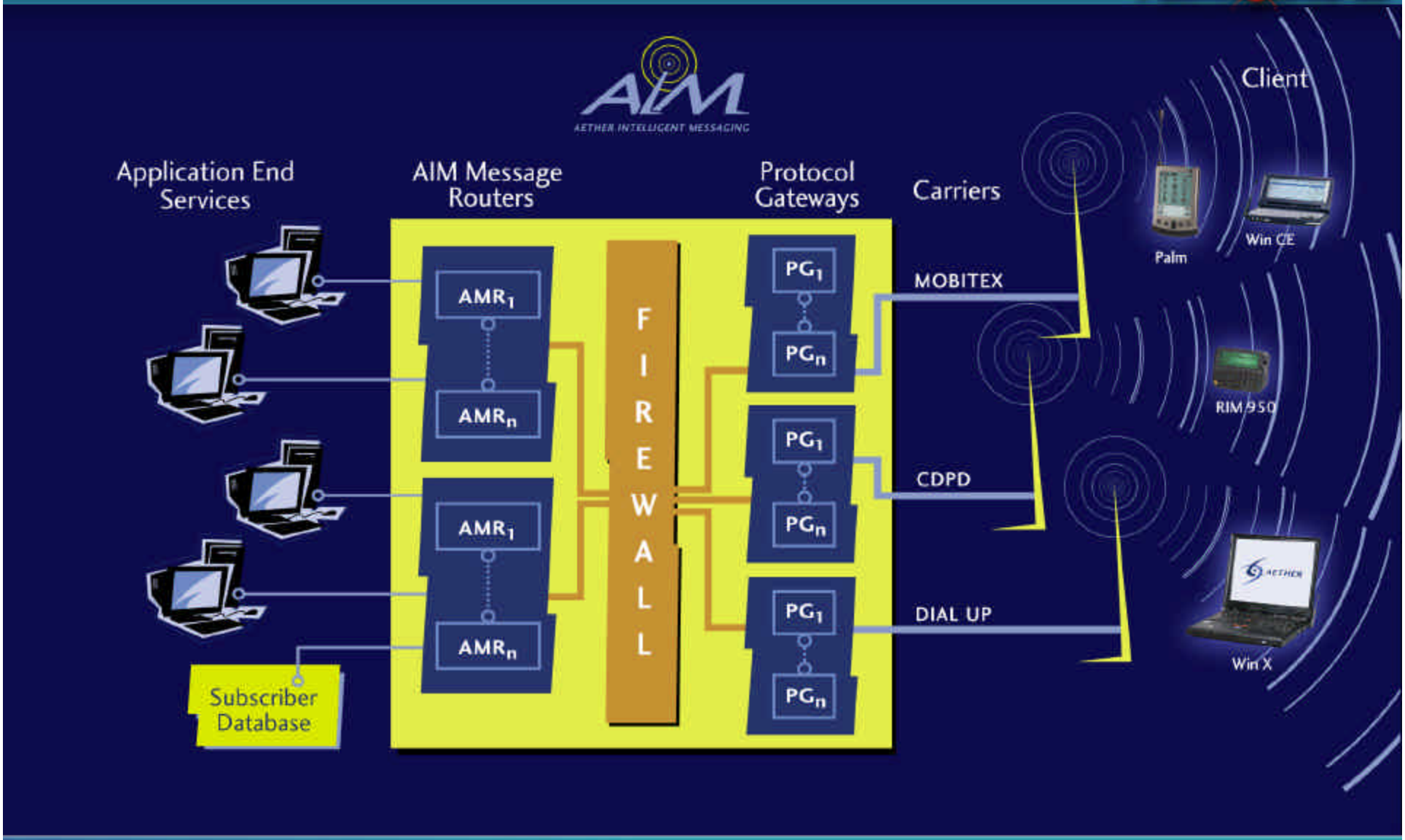
3Com
More connected.

Agenda

- AIM Architecture
- Client SDK for Palm Computing® Platform
- Developing a Wireless Application

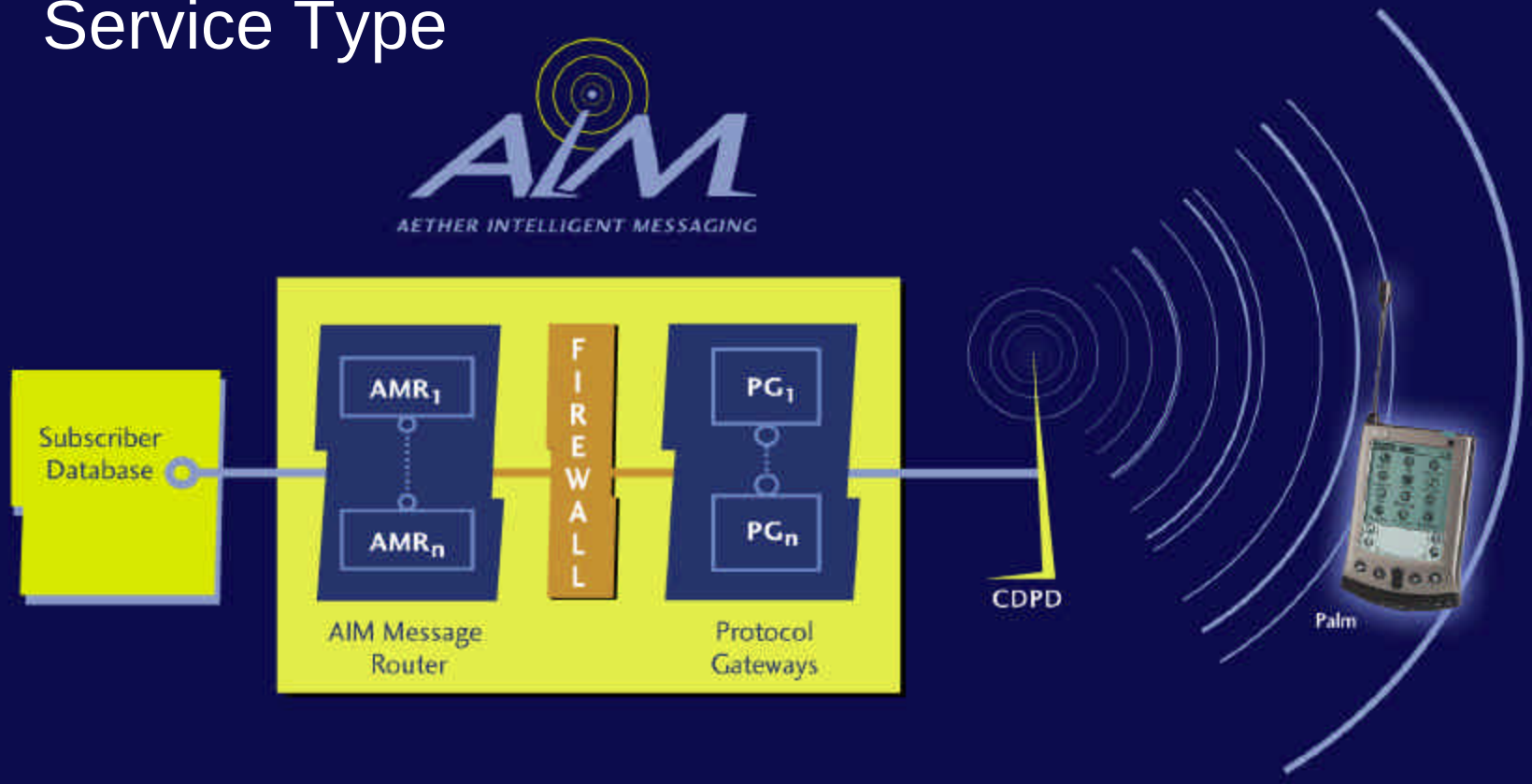


Architectural Overview



Authenticated Level of Service

- Mobile Address and Service Type
- User ID/Password Challenge and Service Type



AIM Message Routing

- Routing info is contained in the AIM Message Header



AIM Message Header

AMR Uses for Routing Messages

- Service type
  - Quote service
- Server ID
  - Directs messages to a specific server for stateful communications
- Message ID
  - Uniquely identifies the message
- Message flags
  - AIM version info, compression, security



Supported Platforms

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- Networks
  - CDPD
  - Mobitex
  - ISP Dial-up
  - 802.11
  - Wired LANs
- Platforms
  - Palm (3Com, Handspring, IBM, Qualcomm, Symbol)
  - CE 2.0/2.11 and Windows 95/98/NT
  - RIM 950 and Blackberry Pagers

AIM Client SDK

- Multiple Platforms
- Multiple Networks
- Core Set of APIs
  - I/O Streaming Services
  - Compression Services
  - Transport Services
- Extended Set of APIs
  - Security Services

I/O Streaming Services

- Set of APIs to assist developers streaming in/out application messages
- Supports the common data types supported by Palm OS® software
- Managed big-endian/little-endian issues
- APIs implemented in shared library

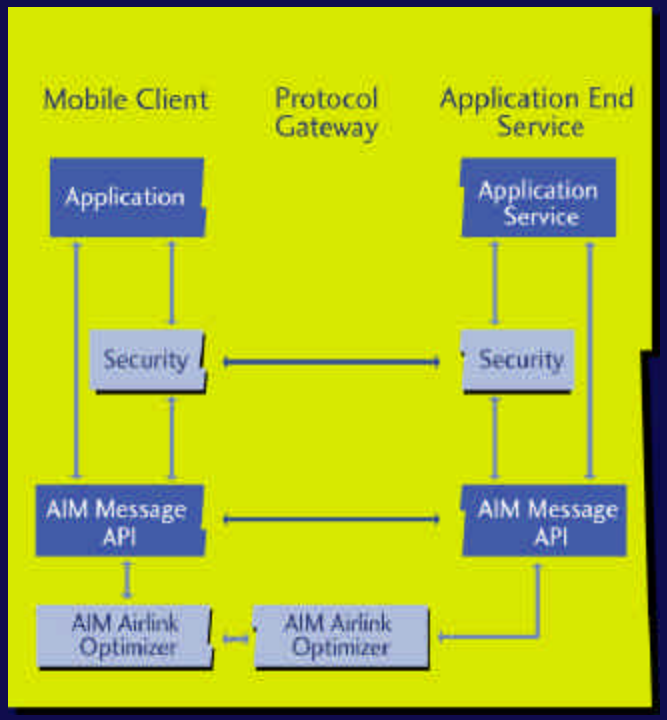
Compression Services

- Set of APIs to assist developers with encoding and decoding application messages
- Based upon Adaptive-Huffman algorithm
- Service provider interface
- APIs implemented in shared library

Transport Services

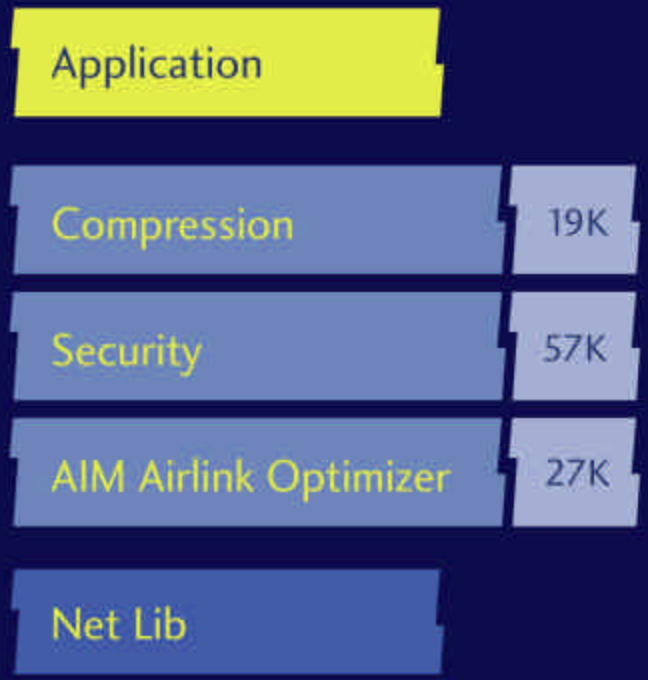
- Reliable, optimized data communication
- Encapsulate the network communications protocol
- Message segmentation/reassembly
- Message duplicate detection
- Message retries/message time-outs
- APIs implemented in shared library

Security Services



- Elliptic Curve Cryptography
  - Licensed from Certicom
- End-to-End Security—Client to Service
- Equivalent to RSA 1024-bit
- Encryption
- Authentication
- Non-repudiation
- Message-based encryption not connection based
- Service provider interface
- APIs implemented in shared library

Palm Implementation



- Metrowerks CodeWarrior v5.0
- Palm OS® Emulator
- AIM Message Wizard
- Five (5) primary APIs
- Three (3) distinct shared software libraries

Developing a Wireless Application

- Market Index Monitor
  - Obtains real-time market indexes for…
    - Dow Jones
    - S&P 500
    - NASDAQ
    - Wilshire
  - Showcases core APIs
  - Runs in Palm Emulator

AeTransportOpen()

This function opens and initializes the AIM Transport services. This function must complete successfully prior to calling any of the other Transport APIs.

```
Err AeTransportOpen(UInt              libRefNum,
                    NETWORK_PROTOCOL networkProtocol,
                    UShort            reserved);
```

AeTransportCreateContext()

This function creates a communication endpoint to use for sending and receiving application messages.

```
Err AeTransportCreateContext(UInt         libRefNum,
                             HCCONTEXT*   hCommContext,
                             UShort       applID);
```

AeTransportSend()

This function sends an application message to the target destination specified by the LinkStationID.

```
Err AeTransportSend(UInt              libRefNum,
                    HCCONTEXT         hCommContext,
                    LINKSTATION_ID    hLinkStationID,
                    Short             msgFlags,
                    UShort*           msgID,
                    VoidPtr           msgPtr,
                    UInt              msgLen);
```

AeTransportReceive()

This function receives a message from a communicating AES.

```
Err AeTransportReceive(UInt            libRefNum,
                       HCCONTEXT       hCommContext,
                       UShort*         msgID,
                       AIM_MSGTYPE*    msgType,
                       VoidHand*       hMsg,
                       UInt*           msgLen);
```

AeTransportIdle()

This function gives the Transport Library an opportunity to perform background tasks. For example, message retries.

```
Err AeTransportIdle (UInt libRefNum);
```

AeTransportCloseContext()

This function closes a previously created communication endpoint.

```
Err AeTransportCloseContext(UInt         libRefNum,
                            HCCONTEXT    hCommContext);
```

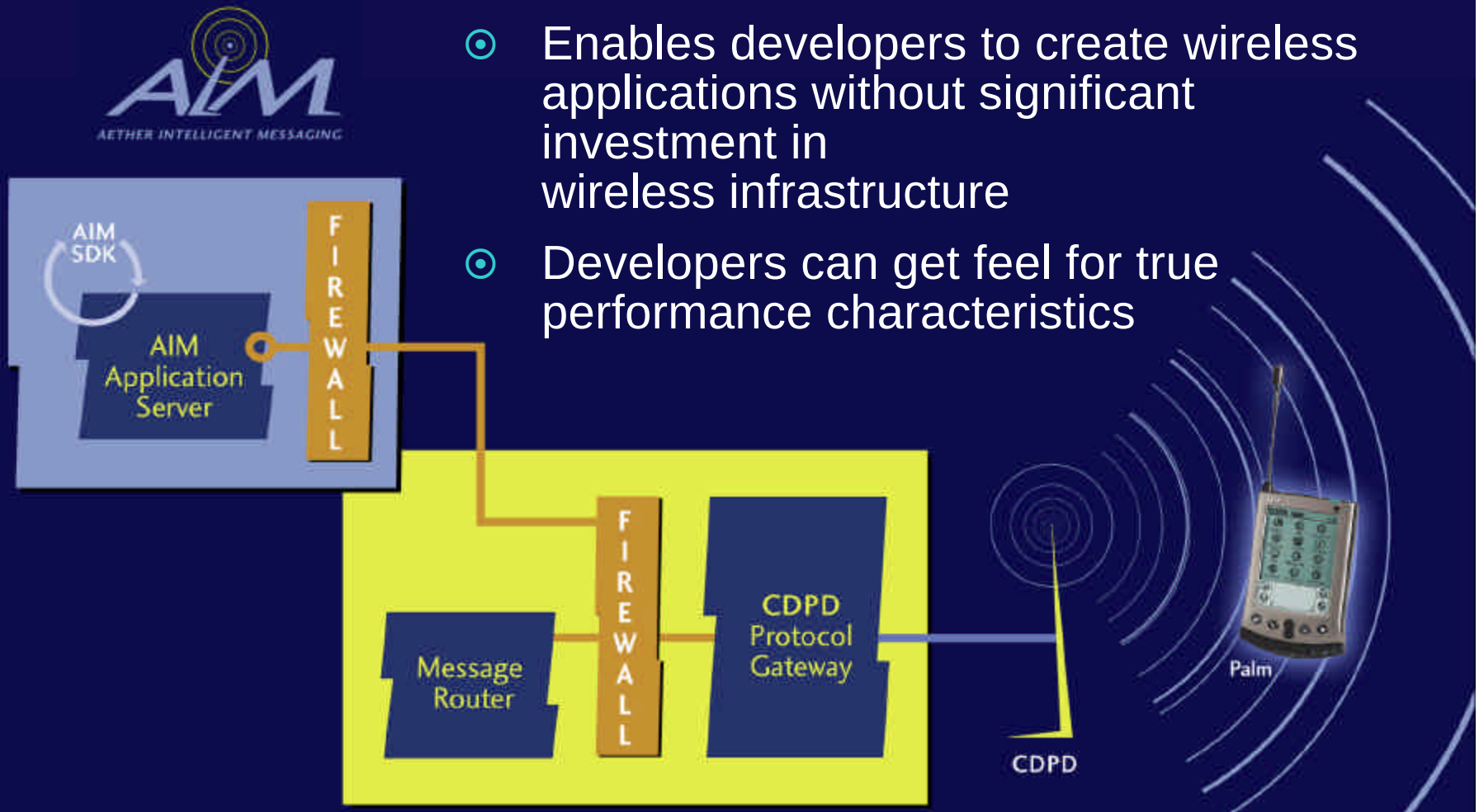
AeTransportClose()

This function closes and frees the resources associated with the AIM Transport services.

```
Err AeTransportClose (UInt        libRefNum,
                      Boolean     bImmediate);
```

AIM Developer's Network

- Enables developers to create wireless applications without significant investment in wireless infrastructure
- Developers can get feel for true performance characteristics



What's Next?

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- Platforms
  - EPOC
  - Others
- Other networks
  - Ricochet II
  - ReFlex
- Advanced Compression
- Customers
  - Reuters
  - Discover
  - Schwab
  - OpenSky
  - Merrill Lynch
  - Bear Stearns
  - RiverBed Technologies
  - Oracle
  - Many others to be announced…

Q&A

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Thank You!

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